# International Trade: theory and applications

lecture notes

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# What we will cover

- Dual representation of general equilibrium
- Standard competitive models
- Imperfect competition, scale, agglomeration, and competition policy
- Multinational enterprise and offshoring
- Political economy of trade and trade policy
  - Tariffs
  - Regulation of firms (competition policy vis-à-vis MNEs, and also tax competition)
  - Regional integration
- Empirical methods (gravity models, trade and wages, testing political economy determinants of policy)

# Approach and literature

- Where appropriate, we will be working with a dual representation of trade under general equilibrium. This means we will often be relying on the basic structure of Dixit and Norman (1980/86). This will be applied to current topics (competition, politics, inequality, geography).
- The course page includes a digital reader, covering textbook readings and articles. This includes topics we will not be covering, but that you really should be aware of. You will get a copy at the end of the course, on CD

# General equilibrium with expenditure and revenue functions

# Some notation and definitions

- $x$ represents vector of final consumption quantities
- $v$ represents vector of primary factor endowments
- $p$ represents the vector of final good quantities
- $\omega$ represents the vector of primary factor prices
- $u$ represents utility

# The Expenditure Function

- We assume that a "representative consumer" has a utility function and faces a budget constraint as follows:

$$u = g(c), \qquad g' > 0$$

$$c = h(x), \qquad \frac{\partial h}{\partial x_i} > 0, \ \frac{\partial^2 h}{\partial x_i \partial x_j} > 0, \ \frac{\partial^2 h}{\partial x_i^{\,2}} \ \forall i \neq j$$

$$y = p \cdot x = \sum_{i=1}^{n} p_i x_i$$

We will drop the representative consumer, but not identical preferences, when we explore trade and inequality. *g*(.) is useful when we explore investment.

# The Expenditure Function

- From the constrained utility maximization problem we can arrive at the following:

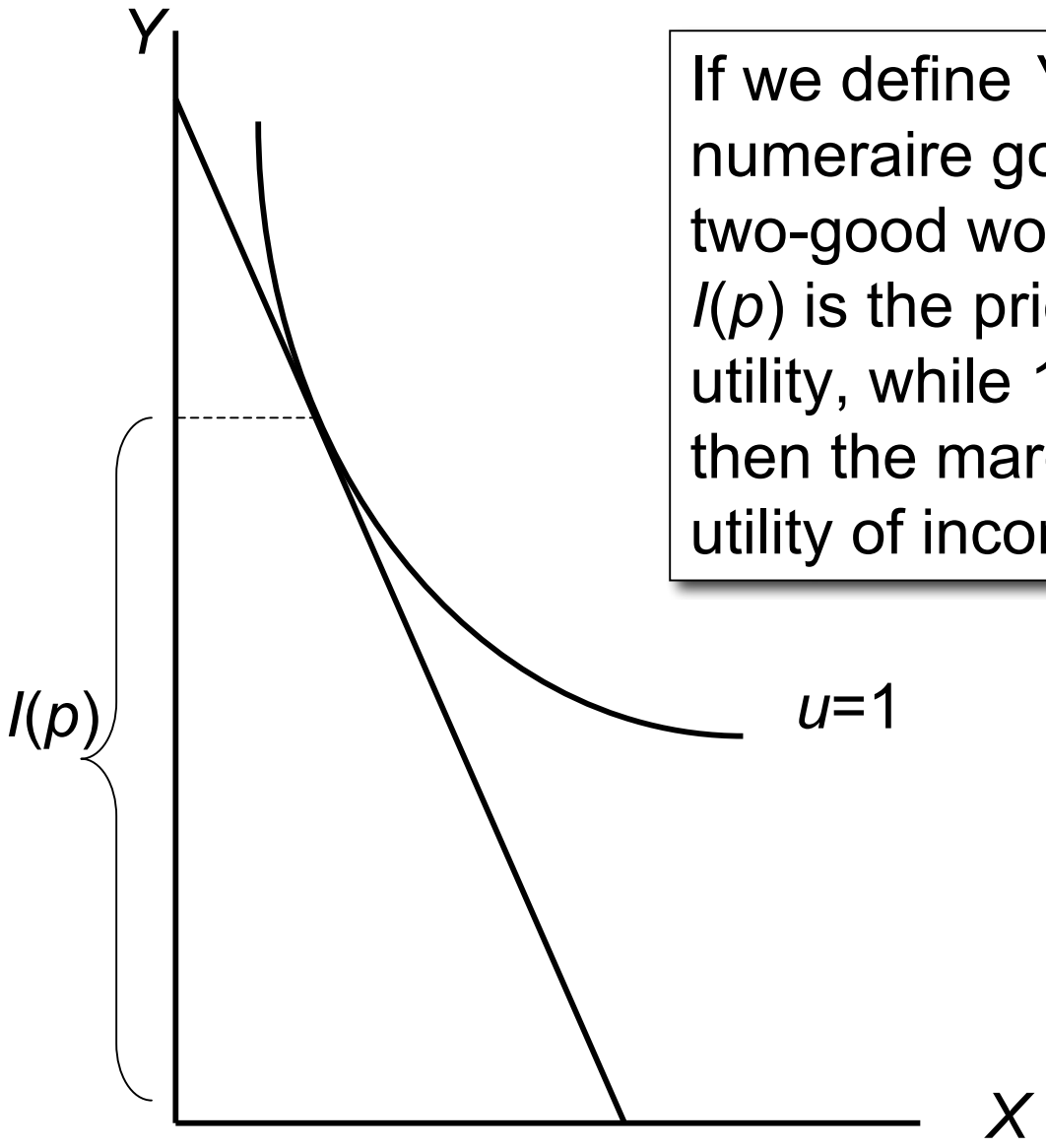
$$L = \max_{x} \left\{ h(x) \middle| p \cdot x \leq y \right\}$$

$$e(p,u) \min_{x} \left\{ p \cdot c \middle| g(h(x)) \geq u \right\}$$

- *e* is essentially the cost function for producing utility from consumption.

Basically, the expenditure function is the minimum cost technology for producing the composite consumption good given preferences.

# The Expenditure Function




If we define *Y* as the numeraire good in a two-good world, then *I*(*p*) is the price of utility, while 1/*I*(*p*) is then the marginal utility of income.

# The Expenditure Function

- The expenditure has some useful properties from the envelope theorem.

$$x_i = \frac{\partial e(p,u)}{\partial p_i}$$

$$e = u \cdot I(p), \qquad I = e(p : u = 1)$$

$$x_i = u \cdot \frac{\partial I(p)}{\partial p_i}$$

- Compensated and uncompensated demands are equal at equilibrium/solution values.

# The Revenue Function

- Turning to firms, we assume convex technologies, and competitive behavior. So, we can represent costs as follows:

$$\beta_j(\omega) = \min_v \left\{ \omega \cdot v^j \,\middle|\, f^j\left(v^j\right) = 1 \right\}$$

$$a_{j,i} = \frac{\partial b_j}{\partial \omega_i} \geq 0$$

- $\beta_j$ is the value of the minimized unit cost function.
- $a_{j,i}$ are unit input coefficients for input i used to produce j.

# The Revenue Function

- We are then able to represent revenue from value added in the economy as follows:

$$r(p,v) = \min_{\omega} \left\{ \omega \cdot v \middle| \beta_j(\omega) \geq p_j \forall j \right\}$$

$$r(p,v) = \max_{x} \left\{ p \cdot x \middle| \sum a_j \cdot x_j \leq v \right\}$$

The revenue function represents the maximization of value added given primary factor supply constraints, or identically the cost minimized value of output where unit costs are at least equal to price and we are limited by our technology set.

# The Revenue Function

- The revenue function also has useful properties:

$$x = \frac{\partial r(p,v)}{\partial p}, \quad \omega = \frac{\partial r(p,v)}{\partial v}$$

$$p \cdot \frac{\partial r(p,v)}{\partial p} = r(p,v) = v \cdot \frac{\partial r(p,v)}{\partial v}$$

From Euler's theorem we get the equivalence of the value added and the final goods output valuation measures of GDP.

# The Closed Economy in General Equilibrium

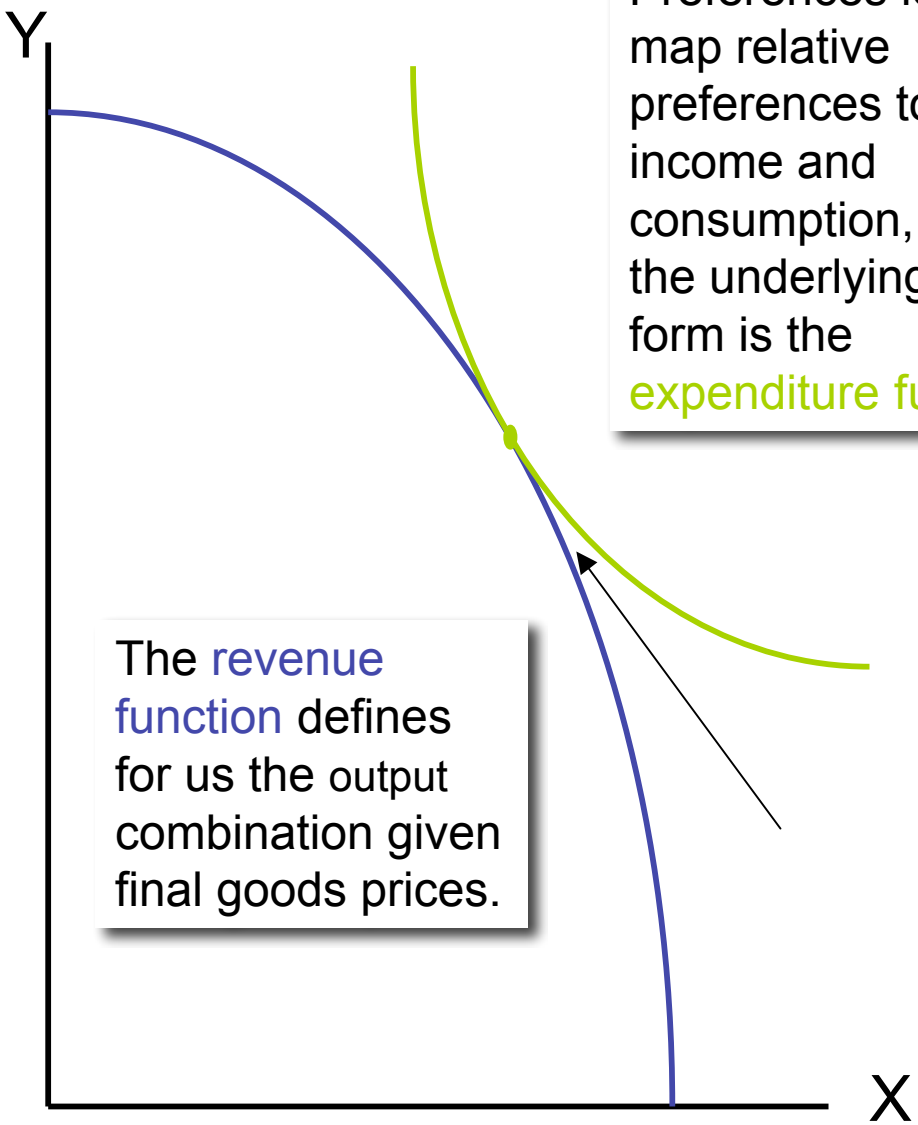
- We need to set expenditures equal to receipts, and clear markets.

$$e(p,u) - r(p,v) = 0$$

$$\frac{\partial e(p,u)}{\partial p} - \frac{\partial r(p,v)}{\partial p} = 0$$

- We can expand this with taxes (which we will do). Also, *R*(.) maps to the macro GDP function, for RBC models and such.

# The closed economy



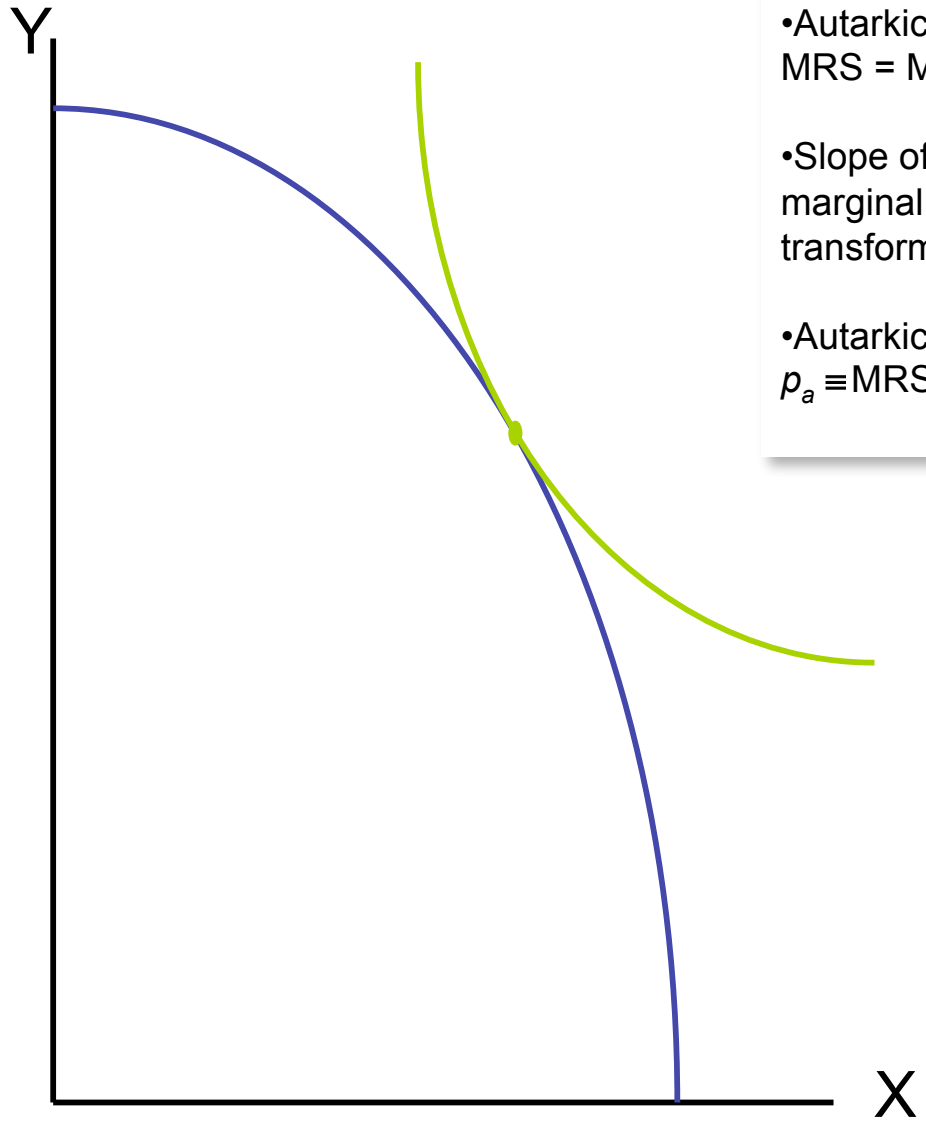

Preferences let us map relative preferences to income and consumption, where the underlying dual form is the expenditure function.

The revenue function defines for us the output combination given final goods prices.

$$e(p,u) - r(p,v) = 0$$

$$\frac{\partial e(p,u)}{\partial p} - \frac{\partial r(p,v)}{\partial p} = 0$$

# The closed economy




- Slope of social indifference curve ≡ marginal rate of substitution (MRS)
- Autarkic equilibrium: MRS = MRT
- Slope of PPF ≡ marginal rate of transformation (MRT)
- Autarkic price, $p_a$ ≡ MRS = MRT

# The Open Economy in General Equilibrium

- We now need to introduce trade: so we need more notation
- *m* will denote imports
- *P** will be used to denote *cif* world prices.
- *b* will represent net capital inflows

# The Open Economy in General Equilibrium

- We can now introduce trade:

Any “extra spending” must be financed externally.

$$e\left(u,P^*\right)=r\left(v,P^*\right)+b$$

$$m-\left(\frac{\partial e\left(u,P^*\right)}{\partial P^*}-\frac{\partial r\left(v,P^*\right)}{\partial P^*}\right)=0$$

Imports are the difference between demand and output values at given internal prices.

# We than have reduced form GE solution values, denoted by ~.

$$\tilde{u} = \tilde{u}\left(P^*, v\right)$$

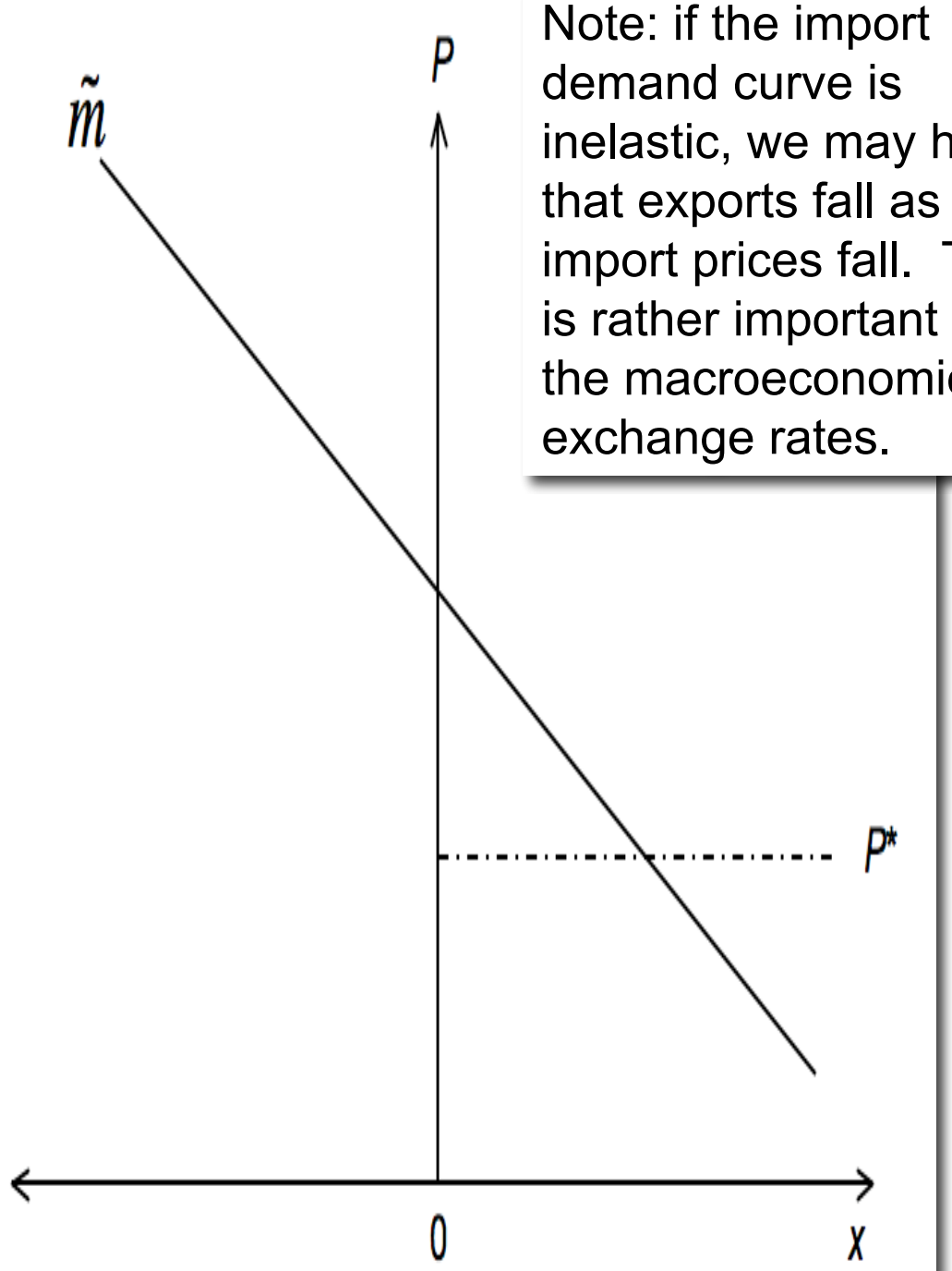
$$\tilde{m} = \tilde{m}\left(P^*, v\right)$$

$$= \frac{\partial e(P^*, \tilde{u})}{\partial P^*} - \frac{\partial r(P^*, v)}{\partial P^*}$$

$$\tilde{\omega} = \tilde{\omega}(P^*, v)$$

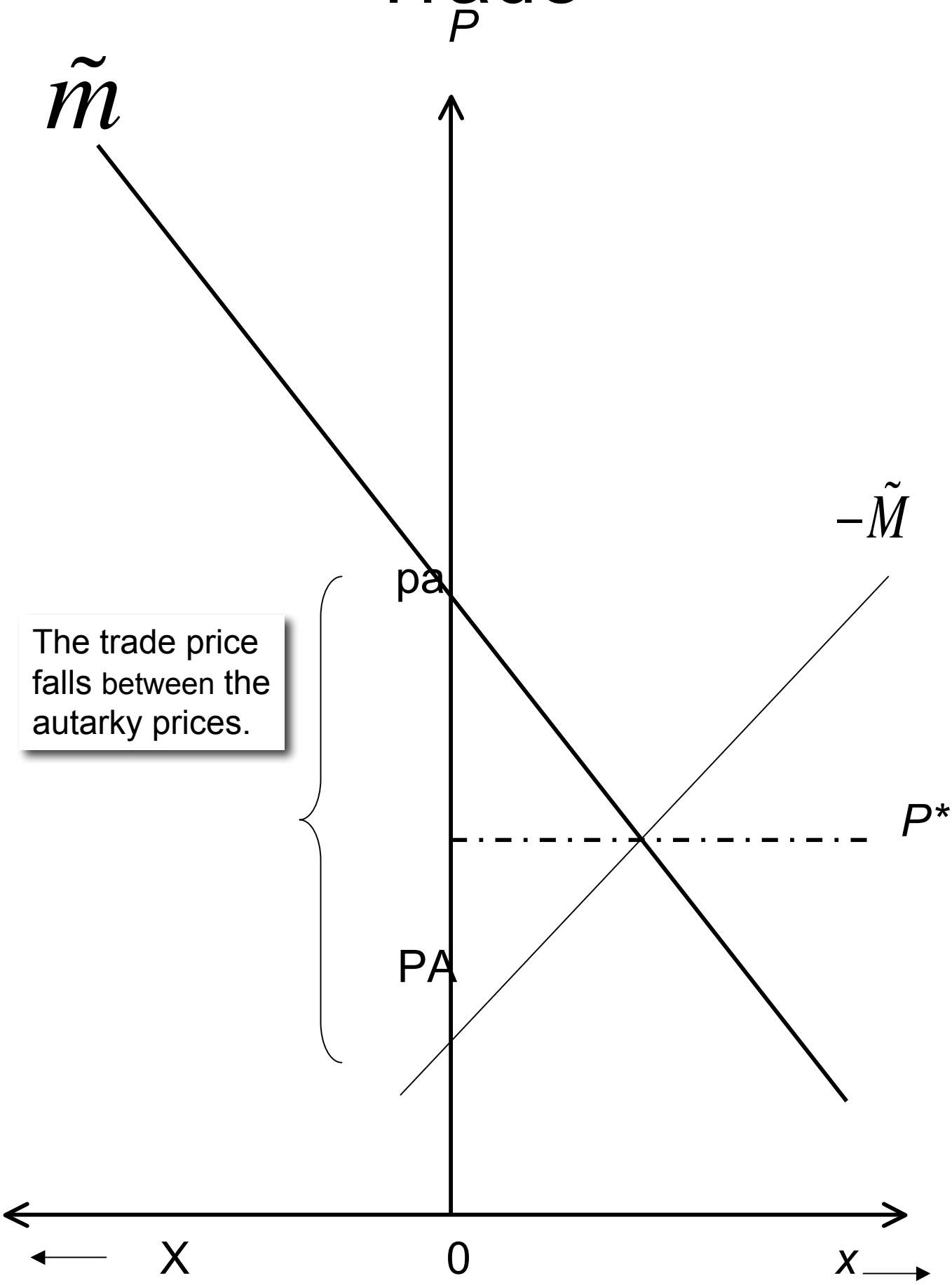
Note: For these values at equilibrium, the compensated and uncompensated quantities must coincide, as we are at equilibrium values, including for utility .

# Graphically, in the 2 good case, we have the following:




Note: if the import demand curve is inelastic, we may have that exports fall as import prices fall. This is rather important for the macroeconomics of exchange rates.

# Trade

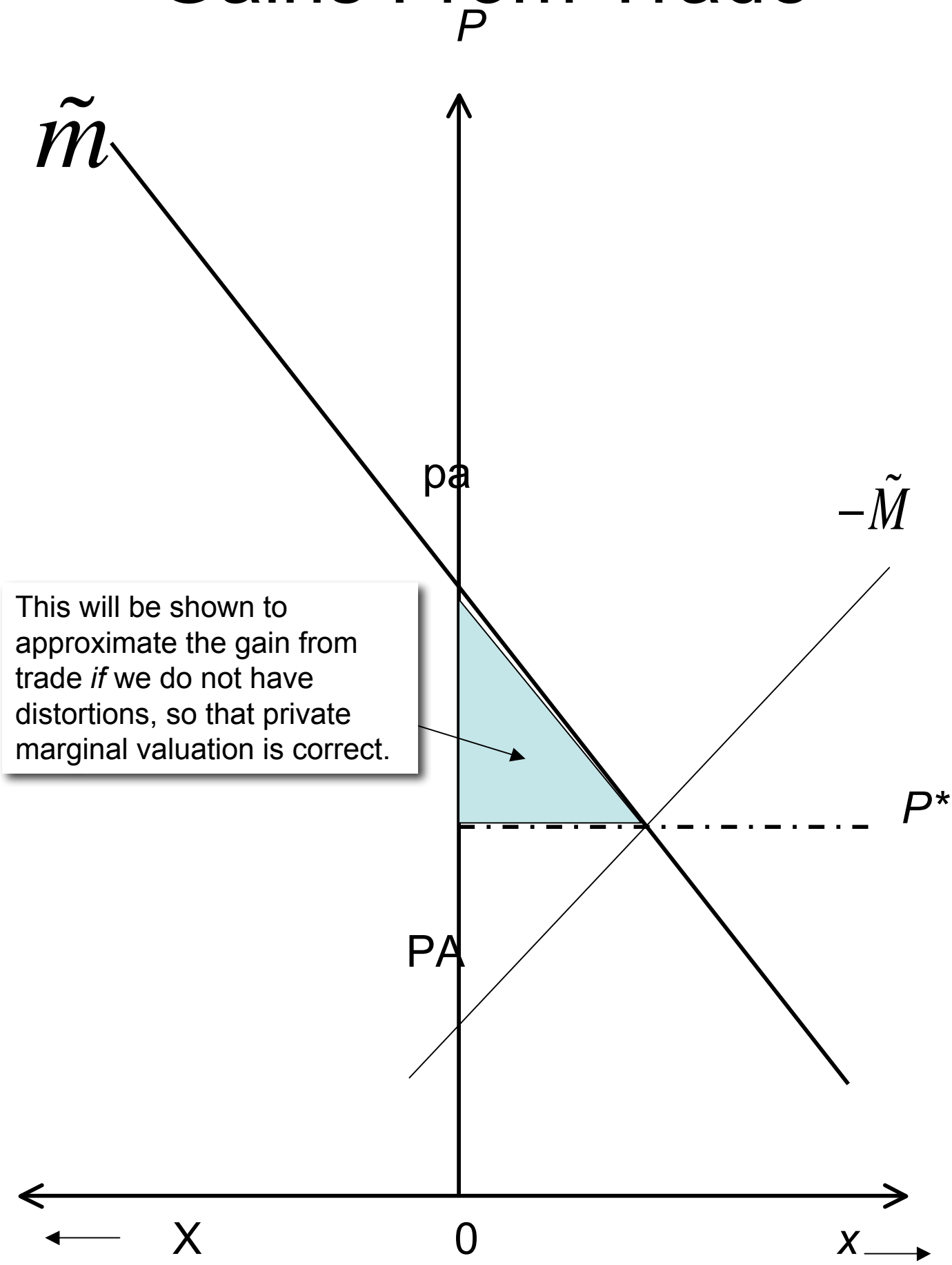

P
$\tilde{m}$
$-\tilde{M}$
pa
The trade price falls between the autarky prices.
P*
PA
X
0
x

# Why would autarky prices be different?

- Technology (Ricardo)
- Relative endowments
  - Capital and labor
  - Fixed factors or natural resources
- Market structure
- Domestic tax structures
- Domestic market size

The reasons for these differences will also determine whether or not we gain from trade.

# Gains From Trade



# The gains from trade

- Start with the balance of trade.

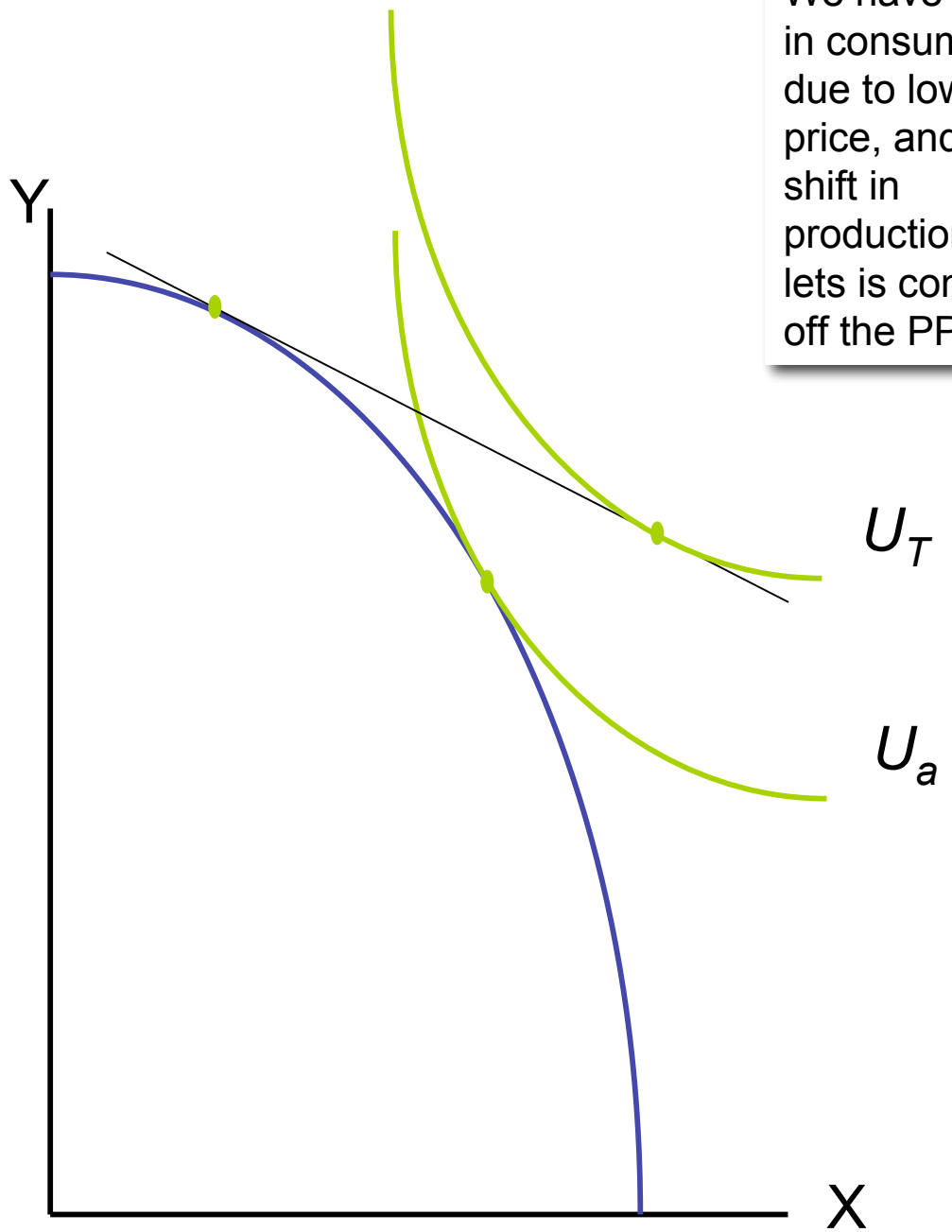
$$b = e(p,u) - r(p,v)$$

- Holding *u* constant we use *b* as a money metric of utility changes. Consider a trade-induced drop in price. Taking a second order Taylor series expansion around a change in price from autarky price $p_a$, we have:

$$\Delta B = -\left(p_a - P^*\right)' \frac{\partial m}{\partial p}\Delta p - \frac{1}{2}\Delta p' \frac{\partial m}{\partial p}\Delta p$$

$$= -\tfrac{1}{2}\frac{\partial m}{\partial p}\Delta p^2$$

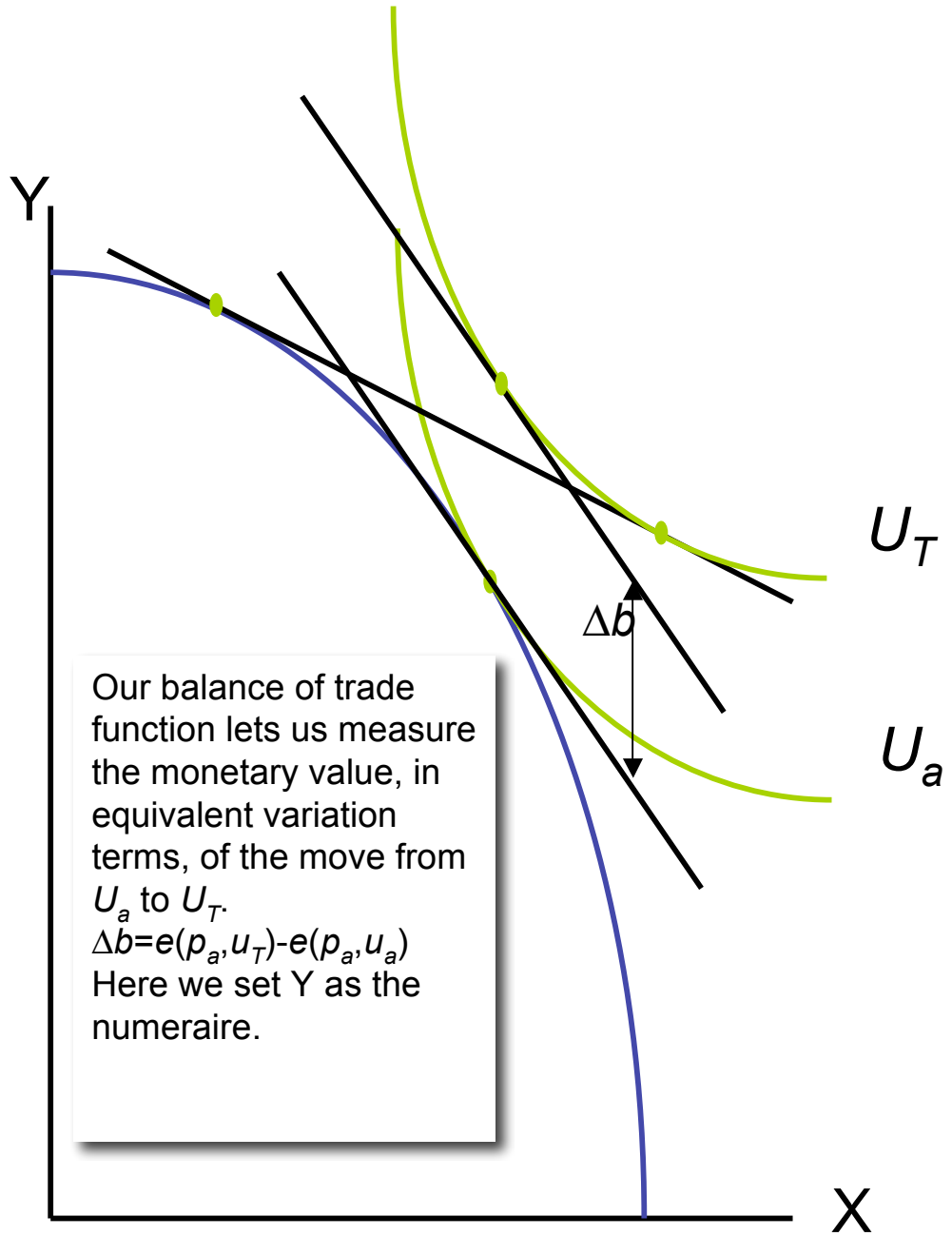
# The Open Economy




$P^* \neg p_a$

We have a shift in consumption due to lower price, and also a shift in production. This lets is consume off the PPF.

# The Open Economy




Our balance of trade function lets us measure the monetary value, in equivalent variation terms, of the move from $U_a$ to $U_T$.
$\Delta b = e(p_a, u_T) - e(p_a, u_a)$
Here we set Y as the numeraire.